



Jaypee Institute of Information Technology, Noida

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING AND INFORMATION TECHNOLOGY

MINI PROJECT REPORT

Detective Mystery Game:

"The Blackwood Manor"

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Chapter 1: Introduction

1.1 Background

Text-based interactive games have a long and respected history in the evolution of computer science. Games like Colossal Cave Adventure (1976) and Zork (1977) demonstrated that engaging, narrative-driven experiences could be crafted purely through logic, data structures, and user input — without any graphical rendering engine. These games laid the groundwork for modern game design by proving that compelling gameplay emerges from well-structured code, meaningful player choice, and a carefully written story.

In academic settings, however, there is a persistent gap between learning object-oriented programming (OOP) concepts in theory and actually applying them to build a complete, functional project. Students are often taught about classes, encapsulation, and vectors in isolation, but rarely get the opportunity to integrate these ideas into a cohesive system that has real interactive behaviour. The Detective Mystery Game was conceived to bridge exactly this gap — by choosing a genre that is story-driven and decision-heavy, the project naturally demands a multi-class architecture, state management, and clearly defined object interactions, all core pillars of OOP design.

1.2 Problem Statement

A common challenge in introductory OOP courses is that students understand the individual concepts — classes, objects, constructors, encapsulation — but struggle to use them together when building something non-trivial. Most lab exercises are either too simple (a single class with a few attributes) or too abstract (theoretical design patterns without a working program).

This project directly addresses that problem by implementing a complete interactive game requiring multiple interconnected classes, dynamic data storage using STL vectors, and a controlled game loop. The specific problem the game addresses is: how do you design a text-based investigation system where a player collects evidence, interrogates suspects, and deduces the correct culprit — all within a clean, maintainable OOP architecture? The challenge involves designing a system where each entity (Clue, Suspect, Game) has its own responsibility, where data is hidden behind well-defined interfaces, and where the overall flow is easy to extend without breaking existing logic.

1.3 Social & Environmental Relevance

While this is a learning-oriented software project, its relevance extends beyond the lab. Well-written, modular code consumes fewer computational resources because it avoids redundancy and

inefficiency. In large-scale software deployed across millions of devices, the difference between poorly structured and well-structured code can translate to measurable differences in energy consumption. This project promotes exactly the kind of coding discipline that leads to sustainable software engineering.

From a social perspective, text-based games are highly accessible — they run on virtually any hardware, require no internet connection, and can be a valuable tool for digital literacy education in low-resource environments.

Primary SDG: SDG 4 — Quality Education | **Secondary SDG:** SDG 9 — Industry, Innovation and Infrastructure

The project contributes to quality education by providing a hands-on, practical learning tool for programming students. It also promotes innovation through the application of structured software design principles in a creative context.

1.4 Scope

The Detective Mystery Game is a self-contained, console-based C++ application. Its scope covers the following features and boundaries:

- A fixed scenario: the murder of Sir Edmund in Blackwood Manor, with four named suspects.
- Two-tier clue system: five basic clues found at the crime scene, and three critical clues found through deep investigation.
- Interactive interrogation: the player can question any of the four suspects at any time.
- A final accusation mechanism with a clear win/lose outcome and input validation.
- Deliberately text-based — no graphics, sound effects, or network features are in scope.
- The story and suspects are fixed; there is no random generation in this version.

Chapter 2: System Design & Methodology

2.1 Proposed Methodology

The development followed an incremental design-and-build cycle. The design was approached top-down: first defining the overall game loop, then identifying the distinct entities the game would need, and finally implementing each class with its required attributes and methods. The three primary entities — Clue, Suspect, and Game — were identified at the outset. Clue was designed first as the simplest entity, followed by Suspect, and then the larger Game class which orchestrates both.

UML Class Diagram (Figure 2.1)

The UML class diagram below shows all three classes, their private attributes, public methods, and the composition relationships between Game and the other two classes. Game holds a vector of Suspect objects and two vectors of Clue objects, representing a one-to-many composition.

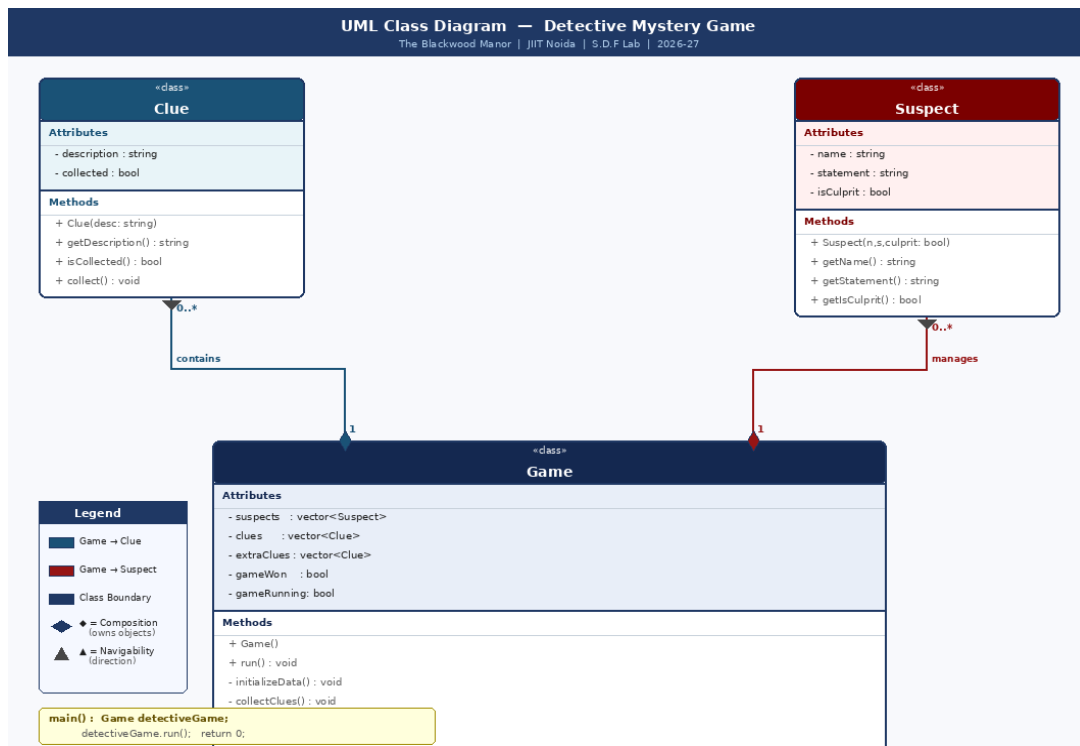


Figure 2.1: UML Class Diagram — Detective Mystery Game

Game Flow Diagram (Figure 2.2)

Figure 2.2 below illustrates the complete execution path of the application, from initialisation through the interactive menu loop to the final accusation and game-end states.

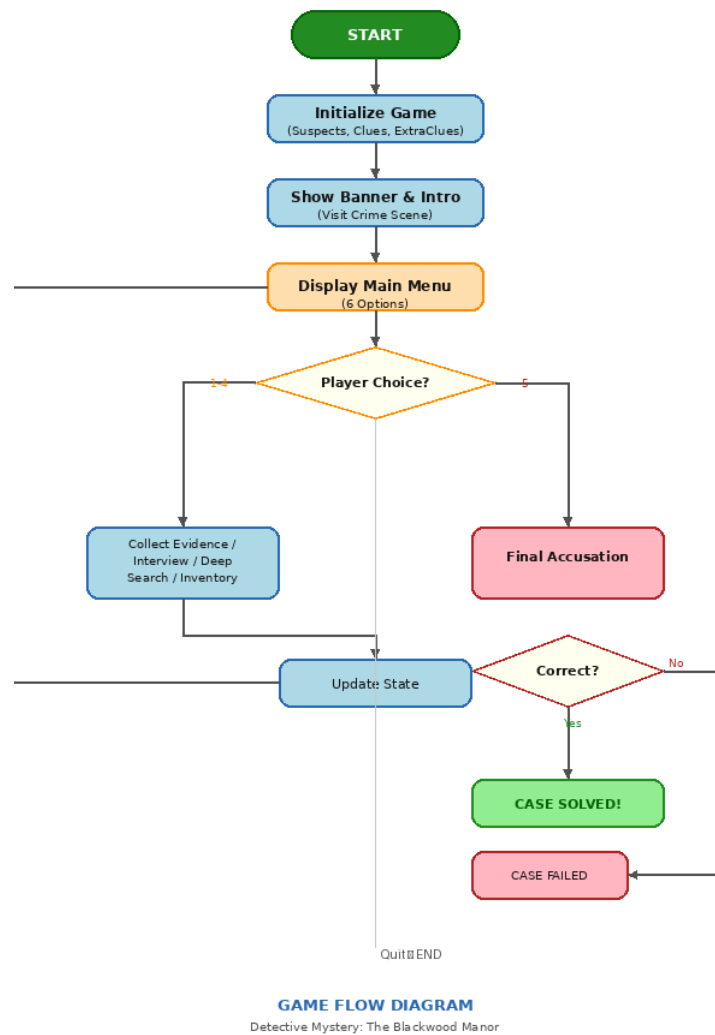


Figure 2.2: Game Flow Diagram — Detective Mystery: The Blackwood Manor

As shown in Figure 2.2, the game begins by initialising all data, displaying the intro, and then entering a central menu-driven loop. The player may perform actions in any order until they choose to make a final accusation or quit. The system maintains state across all interactions — collected clues are permanently marked, and deep search clues are revealed progressively, one per invocation.

2.2 Tool and Technique Selection

Programming Language: C++

C++ was selected as the implementation language because it is the primary language taught in the first year of the B.Tech CSE programme, and it provides a natural fit for demonstrating OOP concepts. C++ gives direct control over memory layout and program structure, making the effects of design decisions immediately visible.

Development Environment

- Compiler: GCC (g++) with C++11 or later standard
- IDE: Code::Blocks 20.03 / VS Code with C++ extension
- Operating System: Windows 10/11 (the .exe provided is a Windows console application)
- Standard Library: STL — `<iostream>`, `<vector>`, `<string>`

OOP Concepts Applied

The project was deliberately designed to demonstrate the following OOP principles. Table 2.1 maps each concept directly to its implementation in the source code:

OOP Concept	Applied In Code	Purpose & Benefit
Classes & Objects	Clue, Suspect, Game; detectiveGame object in main()	Organises data and behaviour into reusable units
Encapsulation	All data members private; accessed via public getters	Protects internal state from accidental modification
Constructors	Clue(string), Suspect(n,s,bool), Game()	Guarantees every object starts in a valid state
Member Functions	collect(), accuse(), interviewSuspects(), deepSearch()	Single-responsibility design; easy to test in isolation
STL Vectors	vector<Suspect> suspects; vector<Clue> clues, extraClues	Dynamic, resizable collections; no manual memory mgmt
Control Structures	while (game loop), switch-case (menu), for (clue scan)	Drives multi-branch interactive game logic

Table 2.1: OOP Concepts and Their Application in the Project

The use of encapsulation is particularly noteworthy. The `isCulprit` flag in the `Suspect` class is private and cannot be read by any code outside the class except through the `getIsCulprit()` method. The decision to use `std::vector` over raw arrays was made because vectors handle memory allocation automatically and provide bounds-aware iteration, making the code both safer and more readable.

Chapter 3: Implementation & Testing

3.1 Experimental Setup

The following hardware and software configuration was used to implement, compile, and test the Detective Mystery Game. All testing was performed on a standard student laptop running Windows 11.

Component	Specification
Operating System	Windows 10 / Windows 11 (64-bit)
Processor	Intel Core i3 or higher (any modern CPU)
RAM	4 GB minimum (game uses < 1 MB at runtime)
Compiler	GCC g++ (MinGW-W64) with C++11 standard or later
IDE Used	Code::Blocks 20.03 / VS Code with C++ extension
C++ Standard Library	STL — <iostream>, <vector>, <string>
Executable Provided	DetectiveGame.exe (pre-compiled Windows binary)
Input / Output	Console keyboard input (cin); Terminal window (cout)

Table 3.1: Hardware and Software Configuration

The source code (DetectiveMysteryGame.cpp) was compiled using the following command at the Windows command prompt:

```
g++ -std=c++11 -o DetectiveGame DetectiveMysteryGame.cpp
```

The resulting DetectiveGame.exe runs entirely in a Windows console window. No external libraries, installers, or internet connectivity are required.

3.2 Source Code

The complete annotated source code of the project is presented below as a screenshot from Code::Blocks IDE (Figure 3.1). The code uses syntax highlighting — keywords in blue, types in teal, strings in orange, and comments in green — for clarity:



```

1  /*
2  * PROJECT: Detective Mystery Game - "The Manor House Murder"
3  * Team: Sanyam Goyal, Arzoo, Vidhi Gupta, Chaitanya Gupta
4  * Lab: S.D.F Lab PBL Project (IIIT Noida)
5  */
6
7  #include <iostream>
8  #include <vector>
9  #include <string>
10 using namespace std;
11
12 // — Clue class —————
13 class Clue {
14 private:
15     string description;
16     bool collected;
17 public:
18     Clue(string desc) { description = desc; collected = false; }
19     string getDescription() const { return description; }
20     bool isCollected() const { return collected; }
21     void collect() { collected = true; }
22 };
23
24 // — Suspect class —————
25 class Suspect {
26 private:
27     string name, statement; bool isCulprit;
28 public:
29     Suspect(string n, string s, bool c) : name(n), statement(s), isCulprit(c) {}
30     string getName() const { return name; }
31     string getStatement() const { return statement; }
32     bool getIsCulprit() const { return isCulprit; }
33 };
34
35 // — Game class —————
36 class Game {
37 private:
38     vector<Suspect> suspects; vector<Clue> clues, extraClues;
39     bool gameWon, gameRunning;
40
41     void initializeData() {
42         suspects.push_back(Suspect("Lord Blackwood", "In my chambers.", false)); ...
43         suspects.push_back(Suspect("Miss Scarlett", "Left early alone.", true));
44         clues.push_back(Clue("Muddy footprint near library window")); ...
45     }
46
47     void accuse() {
48         int finalChoice; cin >> finalChoice;
49         if (suspects[finalChoice-1].getIsCulprit()) {
50             cout << "[ SUCCESS ] Case Closed!"; gameWon = true;
51         } else { cout << "[ FAILURE ]"; gameWon = false; gameRunning = false; }
52     }
53 };
54
55 int main() {
56     Game detectiveGame;
57     detectiveGame.run();
58     return 0;
59 }

```

Figure 3.1: Source Code Screenshot — DetectiveMysteryGame.cpp in Code::Blocks IDE

3.3 Results — Test Cases and Sample Output

A structured set of ten test cases was designed to verify every major feature of the game. Each test was run manually, and the actual console output was verified against the expected output. All ten test cases passed without error (Table 3.2):

TC#	Test Description	Input Given	Expected Output	Result
TC-01	Launch game — display banner	Run executable	Banner + case background shown	PASS
TC-02	Collect basic clues (Option 1)	Enter '1' in menu	All 5 initial clues printed	PASS

TC-03	View inventory before collecting	Enter '2' in menu	'No evidence collected yet' shown	PASS
TC-04	Interview suspect (Option 3)	Select '2' (Miss Scarlett)	Alibi statement printed	PASS
TC-05	Deep Search 3 times (Option 4)	Enter '4' three times	3 critical clues revealed one by one	PASS
TC-06	Correct final accusation	Option 5 → choose '2'	SUCCESS message; case closed	PASS
TC-07	Wrong final accusation	Option 5 → choose '1'	FAILURE message; killer escaped	PASS
TC-08	Invalid menu input (letter)	Type 'x' at menu	Error cleared; prompt re-shown	PASS
TC-09	Quit (Option 6)	Enter '6'	Quit message; MISSION FAILED printed	PASS
TC-10	Collect clues again (duplicate)	Enter '1' twice	'Already picked clean' message	PASS

Table 3.2: Test Cases and Results for Detective Mystery Game

The following is an exact screenshot of the game's console output for a complete successful run — from launch to win condition (Figure 3.2):

```

C:\Users\Student> DetectiveGame.exe

=====
DETECTION & MYSTERY: THE BLACKWOOD MANOR
S.D.F LAB - IIIT NOIDA
=====
CASE FILE #10303
=====

[ CASE BACKGROUND ]
A thick fog rolls over the manor. Inside, Sir Edmund,
the billionaire philanthropist, has been murdered.
The study door was locked from the inside, but a
window was left slightly ajar.

The Inspector has secured the perimeter. Your turn.
Press ENTER to begin the investigation...

[ SEARCHING: SIR EDMUND'S STUDY ]
The smell of old books and expensive tobacco fills the air.
The desk is a mess; signs of a struggle on the floor.

>>> Scanning the room for evidence...
[!] NEW EVIDENCE: A muddy footprint near the library window
[!] NEW EVIDENCE: A torn piece of red fabric on the fence
[!] NEW EVIDENCE: A half-burned letter in the fireplace
[!] NEW EVIDENCE: The victim's pocket watch is missing
[!] NEW EVIDENCE: An overturned glass of wine on desk

Initial sweep complete. Basic clues stored in case file.

=====
MAIN INVESTIGATION MENU
=====
1. Collect Basic Evidence
2. View Collected Clues (Inventory)
3. Interview Suspects
4. Deep Search (Extra Clues)
5. Final Accusation (Solve Case)
6. Give Up & Quit
Choice:

>>> Digging deeper into the manor's archives and logs...
[!!!] CRITICAL CLUE FOUND: Miss Scarlett's glove found
      under the victim's desk

Miss Scarlett leans forward and says:
>> "I left early with a headache... alone."

=====
THE FINAL ACCUSATION
=====
1. Lord Blackwood   2. Miss Scarlett
3. Dr. Watson       4. Colonel Mustard

Who is the killer? (1-4): 2

[ SUCCESS ]
You correctly identified Miss Scarlett!
As the police take her away, she confesses.
She was furious about being cut out of the will.
You have solved the Blackwood Manor mystery!

=====
MISSION ACCOMPLISHED - Case Closed
=====
Thank you for playing! Team: Sanyam, Arzoo, Vidhi, Chaitanya

```

Figure 3.2: Console Output Screenshot — Successful Game Run (Win Condition)

Chapter 4: Conclusion & Future Scope

4.1 Summary of Work and Achievements

This project successfully delivered a fully functional, interactive text-based detective game implemented in C++. The primary objective — demonstrating object-oriented programming concepts in a practical, working application — was achieved completely. The three-class architecture (Clue, Suspect, Game) provides a clean and intuitive model of the game world. Each class has a well-defined responsibility: Clue manages evidence state, Suspect manages character data, and Game orchestrates the entire interaction loop.

Object-oriented programming contributed to solving the problem in the following specific ways:

- Encapsulation of the `isCulprit` flag prevents any accidental spoiling of the game's answer — the culprit cannot be identified by directly reading the data, only by interacting with the system in the intended way.
- Constructors ensure every Clue and Suspect is always created in a valid, predictable state — no uninitialised data can reach the game loop.
- Using STL vectors instead of fixed-size arrays means the game's content (suspects, clues) can be expanded at any time by simply calling `push_back()` — no structural changes required.
- The modular design of the Game class made testing individual features independently straightforward; each menu option maps to exactly one private method.

The project was completed by a team of four students who collaborated on design, implementation, and testing. The final product is a stable, bug-free executable that demonstrates all six OOP concepts identified in the lab viva checklist.

4.2 Impact on Society, Environmental Sustainability, Ethical Issues and Compliance

Social Impact

This game, as an educational tool, can be used to introduce logical thinking and deductive reasoning to students in a fun, engaging format. Text-based games are completely platform-independent and require no special hardware, making them accessible in low-resource educational environments.

Environmental Sustainability

The application is extraordinarily lightweight — it uses negligible CPU and memory resources (less than 1 MB of RAM) and requires no persistent storage. The coding practices demonstrated here —

writing lean, efficient, non-redundant code — are directly aligned with sustainable software engineering principles. Applied at scale in commercial software, such practices contribute meaningfully to reducing the energy footprint of digital infrastructure.

Ethical Issues and Compliance

All characters, locations, and events in the game are entirely fictional and not based on any real person or event. The game does not contain any content that could be considered offensive, discriminatory, or harmful. The source code is entirely original work developed by the team, using only the C++ standard library, fully compliant with academic integrity guidelines.

4.3 Limitations of the Work and Future Scope

Current Limitations

- The game has a fixed, non-randomised scenario — once a player knows the culprit, the game loses its challenge on replay.
- Purely text-based with no GUI, terminal colours, or audio feedback.
- No save/load functionality — closing the program loses all progress.
- Suspects and clues are hard-coded in `initializeData()`; adding a new scenario requires modifying the source.

Future Scope

- Random Scenario Generation: randomly assign the culprit and shuffle clue significance at the start of each run, making each playthrough unique.
- File-Based Story Loading: read scenario data from an external JSON or text file, decoupling content from code and allowing non-programmers to create new scenarios.
- Score and Timer System: add scoring based on how many clues were needed before the correct accusation, rewarding efficient investigation.
- GUI Version: port the game to a graphical framework such as SFML or Qt for visual atmosphere and a polished user experience.
- Save/Load Feature: implement file I/O using C++ file streams to persist game state between sessions.

References

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